Heavy Water Electrolysis with Nickel or Palladium Electrodes as a Low-Energy Proton–Electron Capture Initiator: Connection to LENR and the Fleischmann–Pons Effect

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Abstract

We propose a theoretical connection between the Fleischmann–Pons electrolysis experiment and the PECCNAL proton–electron capture mechanism. Electrolysis of heavy water (D_2O) using nickel or palladium electrodes drives deuterium ions (D^+) into the metal lattice, where geometric confinement reduces the proton–electron separation to femtometre scales. We argue this enables $p + e^- \to n$ reactions without external energy input, as the metal crystal provides the required alignment. Produced neutrons initiate the PECCNAL chain, generating excess heat detectable via precision calorimetry. This framework provides a theoretical basis for the anomalous heat observed in LENR experiments.

Keywords: LENR, cold fusion, heavy water, electrolysis, palladium, nickel, Fleischmann-Pons, calorimetry

1 Introduction

In 1989, Fleischmann and Pons [?] reported anomalous excess heat during electrolysis of D_2O using a palladium cathode. The result was controversial and poorly reproducible, yet some laboratories reported consistent excess heat. The Widom–Larsen theory [?] proposed that surface proton–electron capture ($p + e^- \to n$) could explain LENR phenomena.

We extend this framework using the PECCNAL mechanism and propose that nickel — far cheaper than palladium — may serve as an effective electrode material.

2 Metal Lattice as Geometric Aligner

In a metal crystal, deuterium atoms occupy interstitial sites separated by $\sim$0.1–0.3 nm. The metal lattice:

1. Confines D atoms at fixed positions

2. Reduces D–D and D–e$^-$ distances to near-nuclear scales
3. Provides conduction electrons as capture targets
4. Eliminates the need for external proton acceleration

The conduction electron density in nickel is:

$$n_e = \frac{\rho N_A Z_{\text{valence}}}{M} = \frac{8908 \times 6.022 \times 10^{23} \times 2}{58.69} = 1.83 \times 10^{29}\,\text{m}^{-3} \tag{1}$$

This is 10^{12} times higher than hydrogen gas at STP, dramatically increasing capture probability.

3 Electrolysis Setup

Table 1: Proposed electrolysis configuration

Parameter	**Value**
Cathode	Nickel or palladium plate/wire
Anode	Stainless steel
Electrolyte	D_2O + dissolved Li_2CO_3
Voltage	3–9 V
Current	0.1–2 A
Temperature	Room temperature

Li_2CO_3 serves dual purpose: electrolyte (conductivity) and Li-6 source (neutron capture for energy extraction).

4 Reaction Sequence

$$\mathrm{D}^+ + e^-_{(\mathrm{Ni})} \to n + \nu_e \tag{E1}$$

$$n + {}^2\mathrm{H} \to {}^3\mathrm{H} + p + 2.22\,\mathrm{MeV} \tag{E2}$$

$$n + {}^6\mathrm{Li} \to {}^4\mathrm{He} + {}^3\mathrm{H} + 4.78\,\mathrm{MeV} \tag{E3}$$

5 Experimental Signature

Excess heat detection via calorimetry:

$$Q_{\text{excess}} = Q_{\text{measured}} - Q_{\text{electrolysis}} = mc\Delta T - VIt \tag{2}$$

If $Q_{\text{excess}} > 0$ and reproducible across multiple trials with D_2O but not H_2O, this constitutes evidence for isotope-dependent nuclear reactions consistent with PECCNAL.

Table 2: Electrode material comparison

Property	**Palladium**	**Nickel**
Cost	$\sim$50 \$/g	$\sim$0.02 \$/g
D absorption	Excellent	Good
Conduction e^- density	High	High
LENR reports	Many	Several (Rossi E-Cat)
Availability	Limited	Abundant

6 Nickel vs Palladium

7 Conclusion

Heavy water electrolysis with nickel or palladium electrodes provides a low-energy pathway to initiate PECCNAL reactions without external proton acceleration. The metal lattice acts as a geometric aligner, reducing proton–electron separation to nuclear scales. Dissolved Li_2CO_3 enables simultaneous neutron energy extraction. This experiment is reproducible in a standard laboratory with minimal equipment.

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References

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